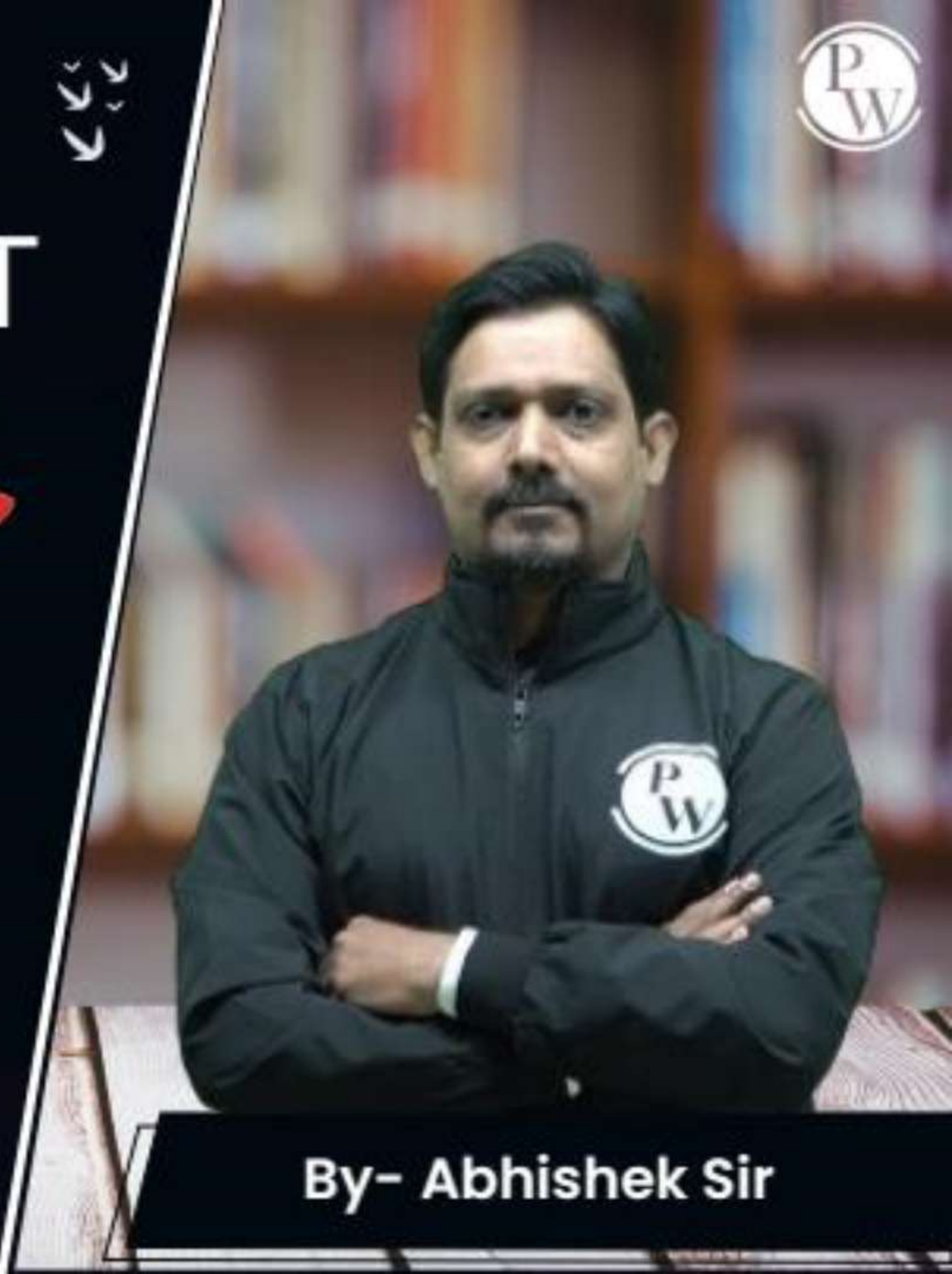
Computer Science & IT



C Programming

Data Types and Operators

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Integer
- Topic — Range of unsigned
- Topic — Range of signed Number
- Topic
- Topic

Slide

Topics to be Covered



- Topic — Problem Solving + C - programming
- Topic — Character
- Topic — Range of Character
- Topic
- Topic

Slide



Topic : Question

00000000 -0

0 - 255

What is the range of 8 bit unsigned Integer

A. 0 to 256

B. 1 to 256

C. 0 to 255

D. 1 to 255

$2^8 - 1 = 256 - 1$

$= 255$

11111111

Slide



Topic : Question

$0 - 2^7 - 1$

0 0 0 0 0 0 0

What is the range of 7 bit unsigned Integer

A. 0 to 128

B. 0 to 127

C. 1 to 128

D. 1 to 127

$0 - 128 - 1$

$0 - 127$

1 1 1 1 1 1 1

$2^7 - 1$

Slide



Topic : Question

-2^{8-1}

-2^{7}

$-128 \longrightarrow 2^{n-1} - 1$

Singed integer

$\rightarrow 2^{7} - 1$

$128 - 1$

$= 127$

What is the range of 8 bit signed Integer

A. 0 to 255

B. -128 to 127

C. -127 to 127

D. -126 to 128

$-2^{8-1} \longleftrightarrow 2^{8-1} - 1$

8 bits

Slide



Topic : Question

short int a;

Size = 2B

What is the range of 16 bit signed Integer

A. 0 to 65536 ✓

B. -32768 to 32767

C. -32767 to 32767

D. -16384 to 16384



-2^{16-1} —— $2^{16-1}-1$ = 32768 - 1

-2^{15}

-32768 — 32767

Slide



Topic : Question

-2^{n-1} to $2^{n-1}-1$

Signed

What is the range of n bit signed Integer

A. -2^n to 2^{n-1}

B. -2^{n-1} to $2^{n-1}-1$ ✓

C. $-2^n - 1$ to 2^{n-1}

D. $-2^n + 1$ to 2^{n-1}

Negative Side 1 No. extra

Slide



Topic : Question

[B]

$2^n - 1$

What is the range of n bit unsigned Integer

A. 0 to 2^{n-1}

B. 0 to $2^n - 1$

C. 0 to $2^{n-1} + 1$

D. $-2^n + 1$ to 2^{n-1}

000..0 ←

[B]

1111..1

$2^n - 1$



Topic : Question

$1 \times 2^5 = 32$ decimal

5 4 3 2 1 :
1 0 0 0 0 0

What is the decimal value of unsigned 6 bit number

100000

A. 64

B. 32

C. 63

D. 31

1 0 0 0 0 0
5 4 3 2 1 0

$2^5 \times 1 = 32$

Slide



Topic : Question

32 — 100000

What is the decimal value of signed 6 bit number

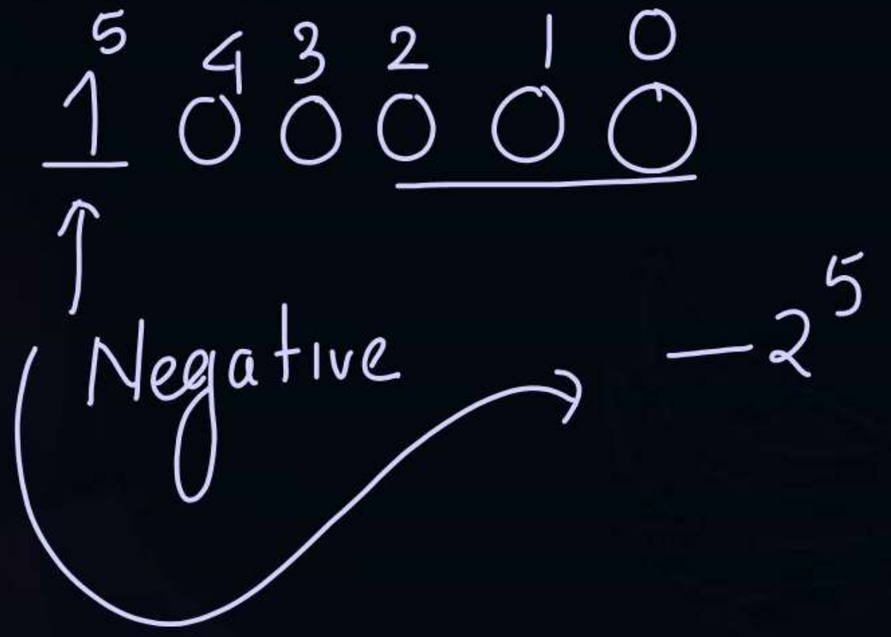
100000

A. -64

B. -32

C. 64

D. 32



Slide

$\overset{3}{1}\overset{2}{0}\overset{1}{0}\overset{0}{0}$ (-8)

-2^3

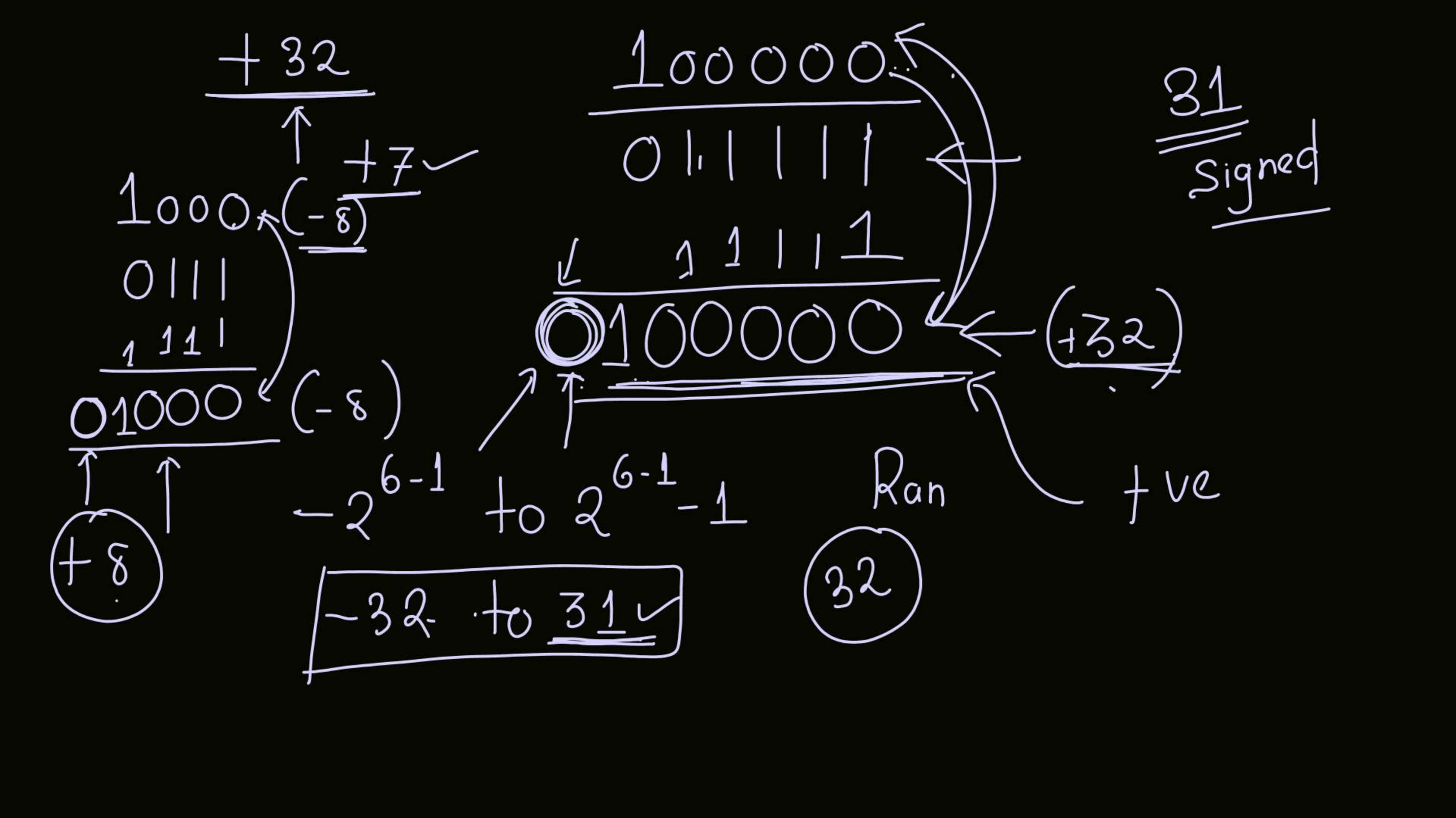
$\underline{100000}$ Signed

MSB – 1 ⟶ Negative No.

-32

$\overset{5}{1}\overset{4}{0}\overset{3}{0}\overset{2}{0}\overset{1}{0}\overset{0}{0}$ (2's complement)

-2^5


+32
1000000
31
Signed
+7
1000
(-8)
0111
1111
01000
(-8)
1 1 1 1 1
0100000
(+32)
+8
-2^{6-1} to $2^{6-1}-1$
Ran
+ve
-32 to 31
32



Topic : Question

unsigned

What is the decimal value of unsigned 6 bit number

100011

A. 64

B. 32

C. 35 ✓

D. 31

1 0 0 0 1 1

↓

32 + 3 = 35

Slide



Topic : Question

100011

What is the decimal value of signed 6 bit number 100011

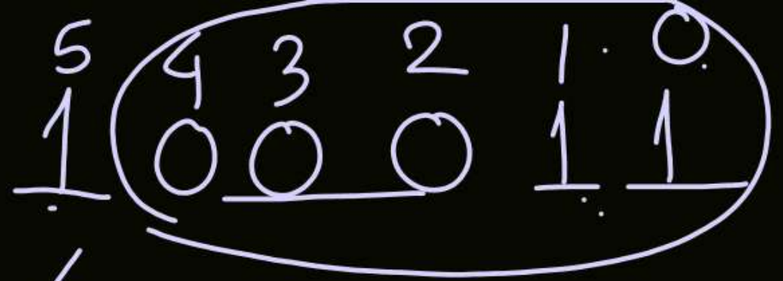
A. 64

B. 29

C. -29

D. 31

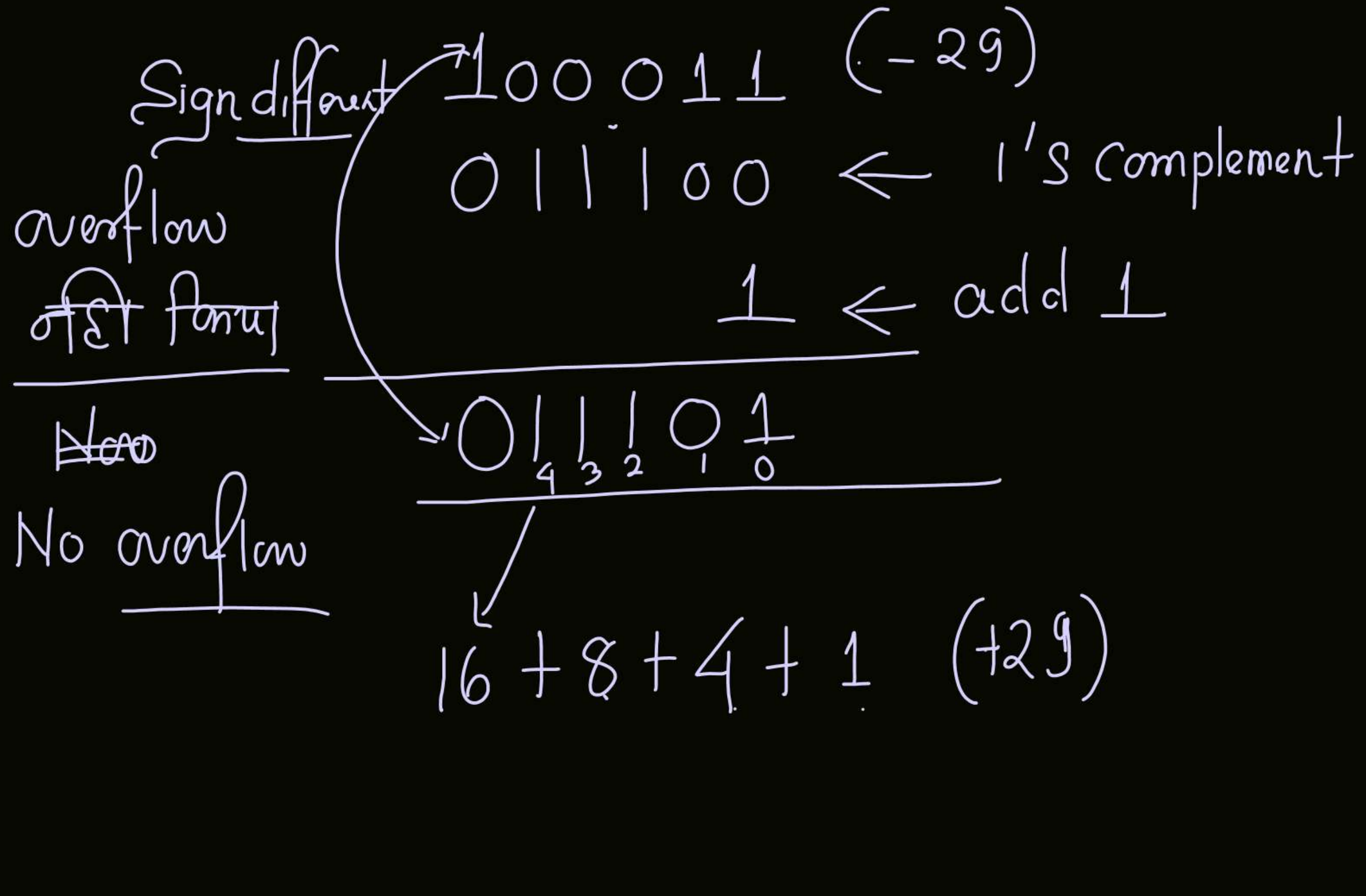
Slide



← -ve No. of

$$-2^5 + 2^1 \times 1 + 2^0 \times 1$$

$$= -32 + 3 = -29 \checkmark$$


Sign diffent
100011 (-29)
011100 ← 1's complement
1 ← add 1
overflow
नहीं flow
No
011101
4 3 2 1 0
No overflow
16 + 8 + 4 + 1 (+29)



Topic : Question

What is the decimal value of unsigned 8 bit number

1000 0011

A. 131

B. 128

C. 259

D. 255

10000011

$2^7 \times 1 + 2^1 \times 1 + 2^0 \times 1 = 131$



Topic : Question

1000 0011 (2's Complement)

$-128 + 3 = -125$

What is the decimal value of signed 8 bit number 1000 0011

A. -125

B. -131

C. 259

D. -259



$-2^7 + 2 + 1 = -128 + 3 = -125$

Slide



First Program

```
#include <stdio.h>
int main()
{
    printf("PW");
    return 0;
}
```

preprocessor directive

#include <stdio.h>

#define ← macro

#include

Standard input/out

Standard function output

$$1000\,0011_{(2)} = 1\times2^{1} + 1\times2^{0}$$

$$2+1=3$$

Library function program execution start

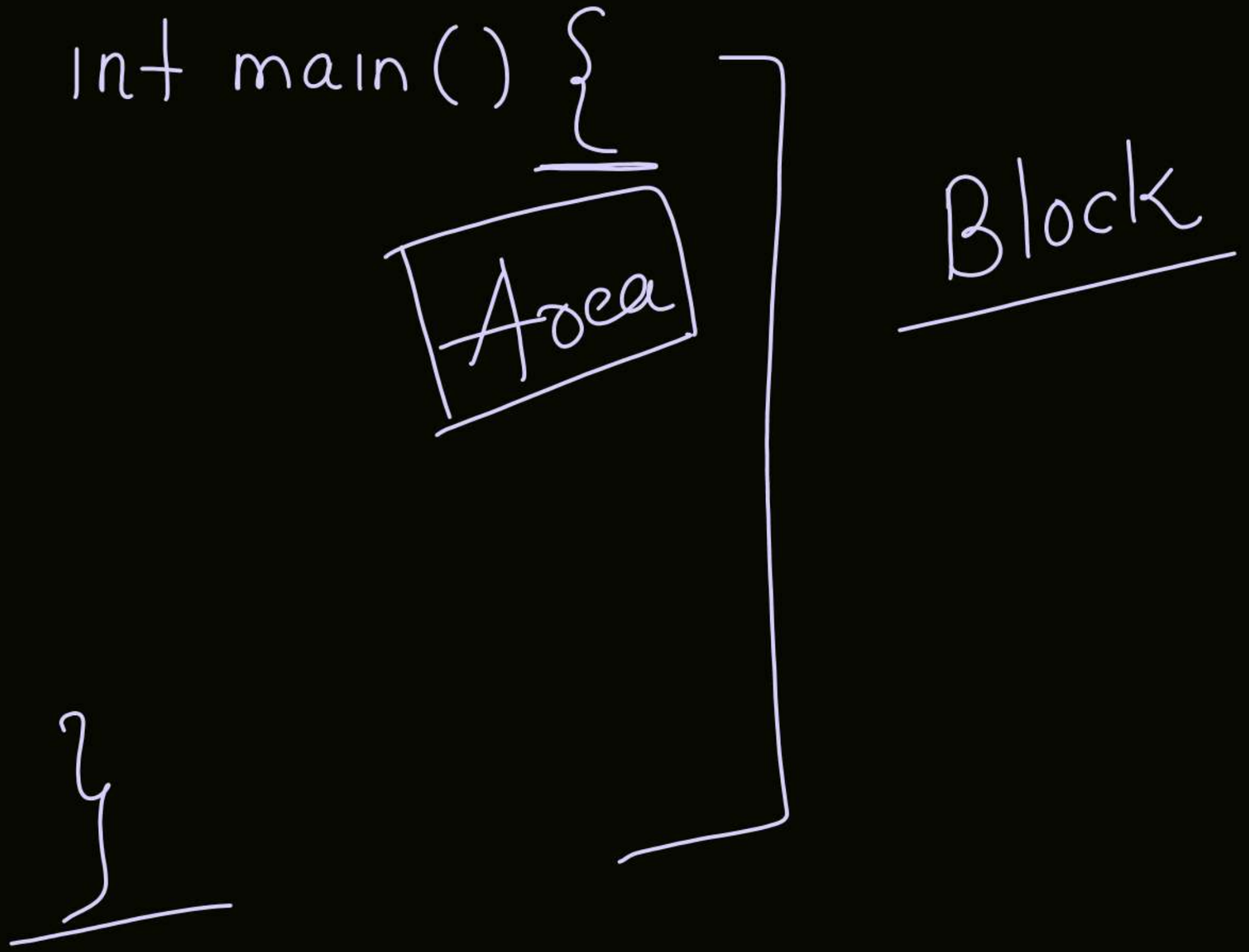
↑

printf( ) first thing execute main()

↑

'C' function

What is function.


int main() {
Area
}
Block

printf(" PW ")

double quote ← string

" "

printf(" My Rank in Gate 2025 willbe 1");

My Rank in Gate 2025 will be 1



Topic : Question

```
#include <stdio.h>
int main(){
        short int a = 10;
        printf("%d", a);
        printf("%u", a);
        return 0;
}
```

2B decimal

"%d" → Sign No

format specifier

unsigned %u

Slide

Integer : "%d" Signed 4B

& "%u" ~~unsigned~~ unsigned

2B

Short int format variable_name %hd

unsigned short int variable %hu



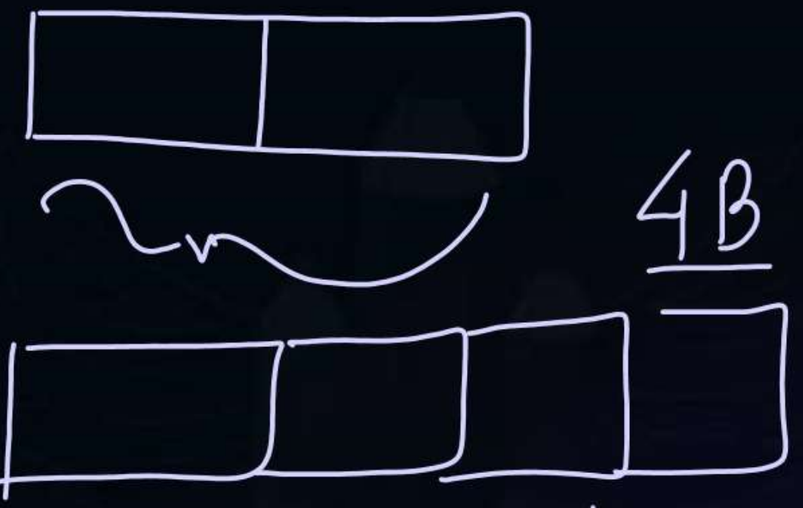
Topic : Question

```c
#include <stdio.h>
int main(){
        short int a = -10;
        printf("%d", a);
        printf("%hu", a);
        return 0;
}
```

Homework

format

%hu : 2B



Slide

Overflow

4 bit No. $-2^{n-1} \longleftrightarrow 2^{n-1}-1$

Range $-8 \longrightarrow +7$

overflow

$$\begin{array}{r} 00010\ (+2) \\ 00110\ (+6) \\ \hline 01000\ (+8) \end{array}$$

$$\begin{array}{r} 2 \rightarrow 0011 \\ 3 \rightarrow 0010 \\ \hline 0101 \end{array}$$



Topic : Question

```
#include <stdio.h>
int main(){
        short int a = 32770;
        printf("%d", a);
        printf("%hu", a);
        return 0;
}
```

Slide



Topic : Question

```
#include <stdio.h>
int main(){
        short int a = 65537;
        printf("%d", a);
        printf("%hu", a);
        return 0;
}
```

Slide



Topic : Reserve keyswords

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU